Lecture 16 recap: Fourier series

- The *Fourier series* of a function $f(x)$ for $(-L, L]$ is

$$f(x) \stackrel{?}{=} \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos k_n x + b_n \sin k_n x \qquad \text{where} \qquad k_n = \frac{n\pi}{L}$$

- The coefficients are given by Euler-Fourier formulae

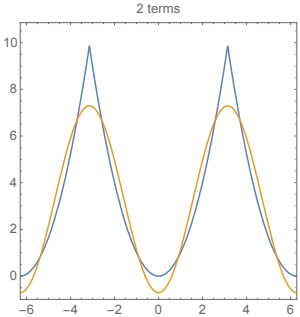
$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos(k_n x)\,\mathrm{d}x, \qquad n = 0, 1, 2, \ldots$$

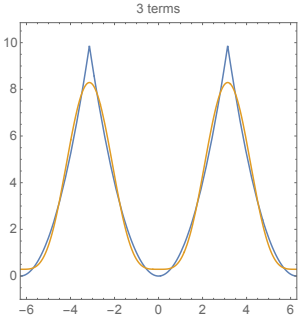
$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin(k_n x)\,\mathrm{d}x, \qquad n = 1, 2, 3, \ldots$$

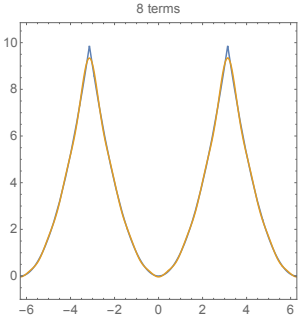
- We denote the periodic extension of $f(x)$ as $F(x)$

Fourier series of $f(x) = x^2$ on $(-\pi, \pi]$

$$F(x) \stackrel{?}{=} \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos(nx)$$

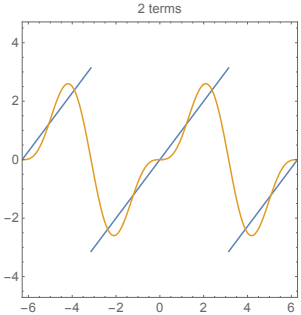


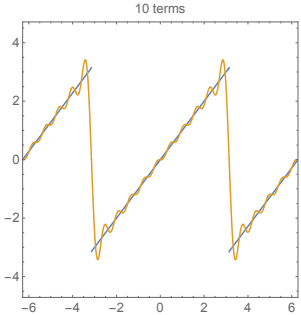


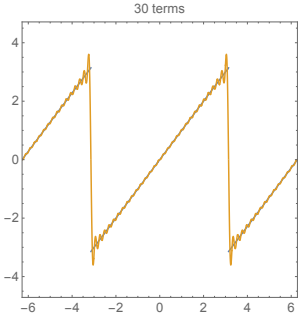


Fourier series of $f(x) = x$ on $(-\pi, \pi]$

$$F(x) \stackrel{?}{=} -2 \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \sin(nx)$$







Lecture 17

In Lecture 17, we will:

1. Outline conditions for pointwise and uniform convergence of Fourier series
2. Have some fun hearing Fourier series (if time permits)

Pointwise convergence of a Fourier series

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$f(x) = x^2$ and $f(x) = x$ piecewise smooth $\Rightarrow$ pointwise convergence of Fourier series

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From Ex. 4.3 + pointwise convergence

$$f(x) = x^2 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}\cos(nx) \quad \Rightarrow \quad f(0) = 0 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$$

$$\Rightarrow 4\sum_{n=1}^{\infty} (-1)^n/n^2 = -\pi^2/3$$

$$\Rightarrow \sum_{n=1}^{\infty} (-1)/n^2 = -\pi^2/12$$

Recap (Topic 2): Uniform convergence of series / Weierstrass test

- The series $\sum_{n=1}^{N} w_n(t)$ converges uniformly to $W(t)$ on D if

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- A uniformly convergent series can be integrated term by term
- But need extra condition term by terdifferentiation

Exercise 4.6: Does the Fourier series of x^2 converge uniformly?

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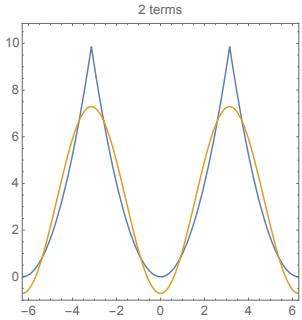
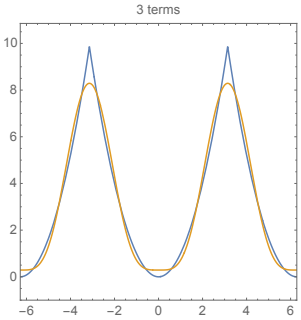
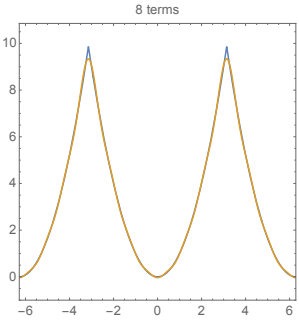
Recall

$$F(x) = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos nx$$







Consider the series $\sum_{n=1}^{\infty} \underbrace{\frac{1}{n^2}}_{a_n}$

$$\sum_{n=1}^{\infty} \underbrace{\frac{(-1)^n}{n^2} \cos nx}_{w_n(x)}$$

Now

$$|w_n| \leq a_n$$

We know $\sum_{n=1}^{\infty} 1/n^2 \quad \left(= \sum_{n=1}^{\infty} a_n\right)$ converges

$\therefore \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos nx$ converges uniformly

$\therefore$ Fourier series of x^2 does also

Theorem

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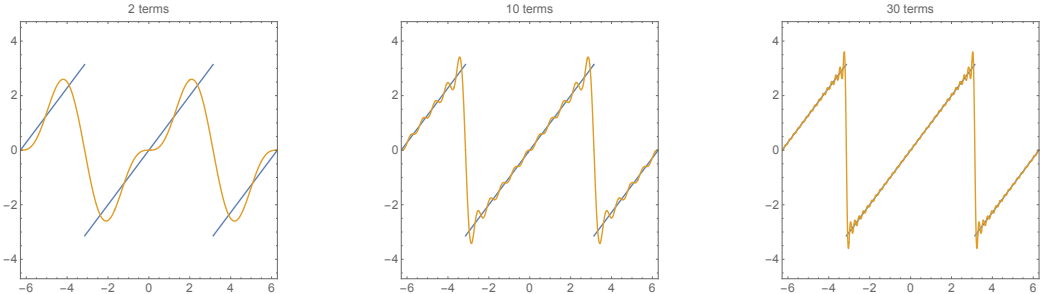

Case study 6: creating an electronic music synthesiser